Recap Checkpoint 1 — after 1.1 (Processors, Input/Output & Storage) + 1.2 (Software)

Name: ____ Date: _ ***Mark:*** ___ / 34

OCR H446 cumulative recap — mixes every topic taught so far. Show all working.

Questions

Q1. State the function of the **Program Counter (PC)** and the **Current Instruction Register (CIR)** during the Fetch–Decode–Execute cycle. **[2]** *(AO1) — 1.1.1*

Q2. A processor uses **pipelining**. (a) Explain what is meant by pipelining. **[2]** *(AO1)*

(b) State **one** situation in which the pipeline may have to be flushed. **[1]** *(AO1)*

[3 total] — *1.1.2*

Q3. A games studio is choosing hardware to render graphics that apply the **same calculation across millions of pixels in parallel**. Explain why a **GPU** is better suited to this than a general-purpose CPU. **[3]** *(AO2) — 1.1.2*

Q4. A digital camera stores photographs onto a removable memory card. (a) Name the **type of secondary storage** used by a typical memory card. **[1]** *(AO1)*

(b) Justify why this storage type is more suitable for a camera than a magnetic hard disk drive, referring to **two** distinct properties. **[2]** *(AO2)*

[3 total] — *1.1.3*

Q5. Explain how an **interrupt** is handled by the operating system, referring to the role of the **interrupt service routine (ISR)** and the **stack**. **[4]** *(AO1) — 1.2.1*

Q6. An operating system uses a **paging** approach to memory management together with **virtual memory**. (a) Explain what is meant by virtual memory. **[2]** *(AO1)*

(b) Describe **one** drawback of relying heavily on virtual memory. **[1]** *(AO2)*

[3 total] — *1.2.1*

Q7. A developer writes a program in a high-level language. (a) State **one** difference between a **compiler** and an **interpreter**. **[2]** *(AO1)*

(b) Place the four stages of compilation in the correct order: *code generation, lexical analysis, syntax analysis, optimisation*. **[2]** *(AO1)*

[4 total] — *1.2.2*

Q8. A software team is choosing between the **Waterfall** and **Agile (e.g. extreme programming)** methodologies for a project whose requirements are expected to change frequently. Recommend a methodology and justify your choice, referring to **two** characteristics of the chosen methodology. **[4]** *(AO2) — 1.2.3*

Q9. Object-oriented programming uses **encapsulation** and **inheritance**. (a) Explain what is meant by **encapsulation**. **[2]** *(AO1)*

(b) Give **one** benefit of **inheritance** to a programming team. **[1]** *(AO2)*

[3 total] — *1.2.4*

Q10. A retailer is replacing its ageing tills. The new tills must run reliably for long shifts, recover quickly from power loss, and let staff carry out card payments and stock look-ups simultaneously. The IT manager must decide on the **type of processor architecture (CISC vs RISC)** and the **type of operating system** (real-time, multi-tasking, distributed). Evaluate suitable choices for **both** decisions, justifying each against the retailer's requirements. **[5]** *(AO3) — synoptic 1.1 + 1.2*